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# Lesson 17
# Delivery Routing Problem

**E211 – Operations Planning II**

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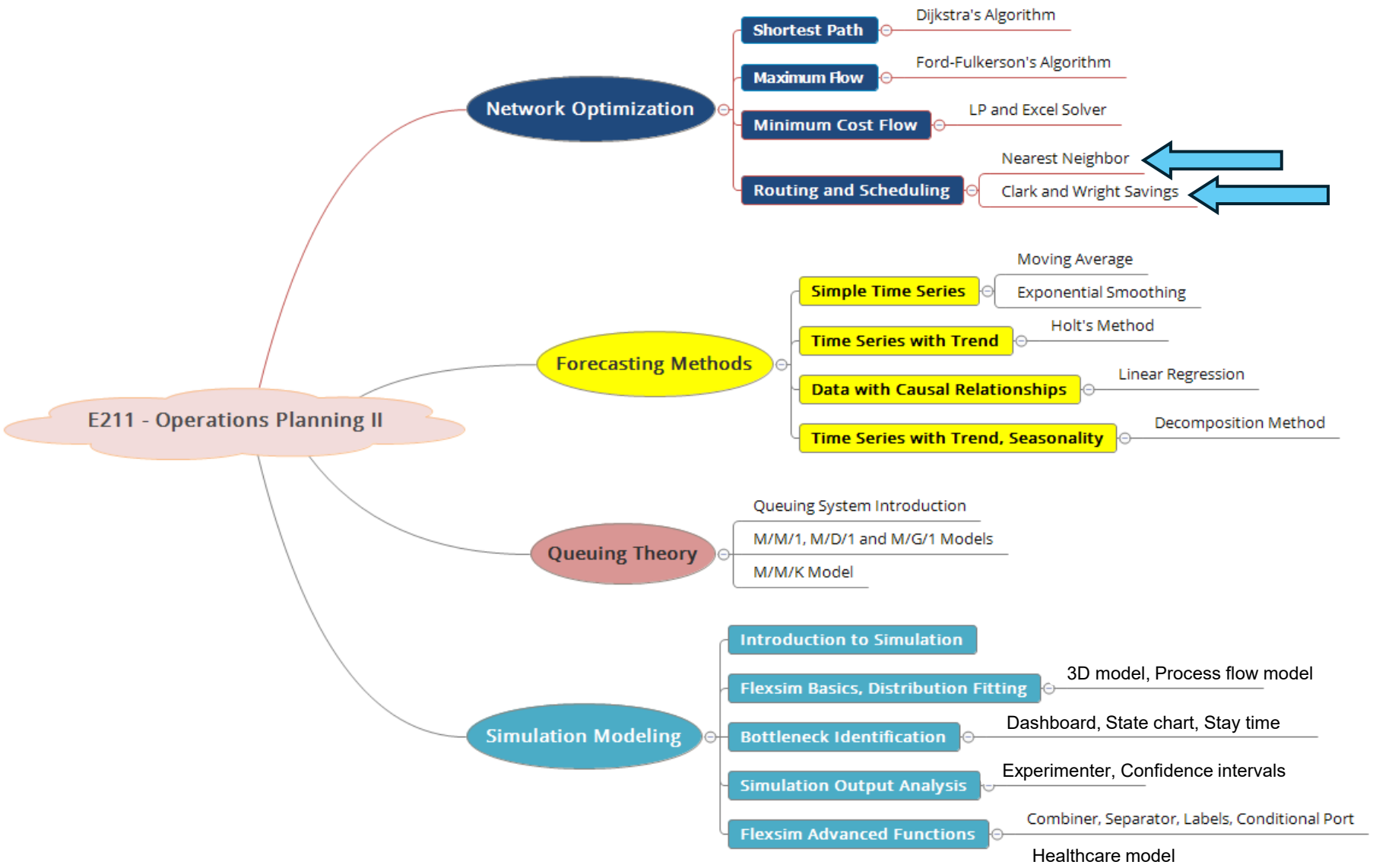
# E211 Topic Tree



# Routing & Scheduling

# Routing and Scheduling Problems

- The scheduling of customer service and the routing of service vehicles are at the heart of many service operations.
  - Service routing: dispatching of installation or service technicians
  - Passenger routing: routing and scheduling of school buses, public buses, ride-sharing
  - Freight routing: pickup and delivery service of logistics firms; postal and parcel delivery
- **Routing** involves forming **the sequence** in which locations are to be visited.
- **Scheduling** involves planning **the timing** for each location to be visited.

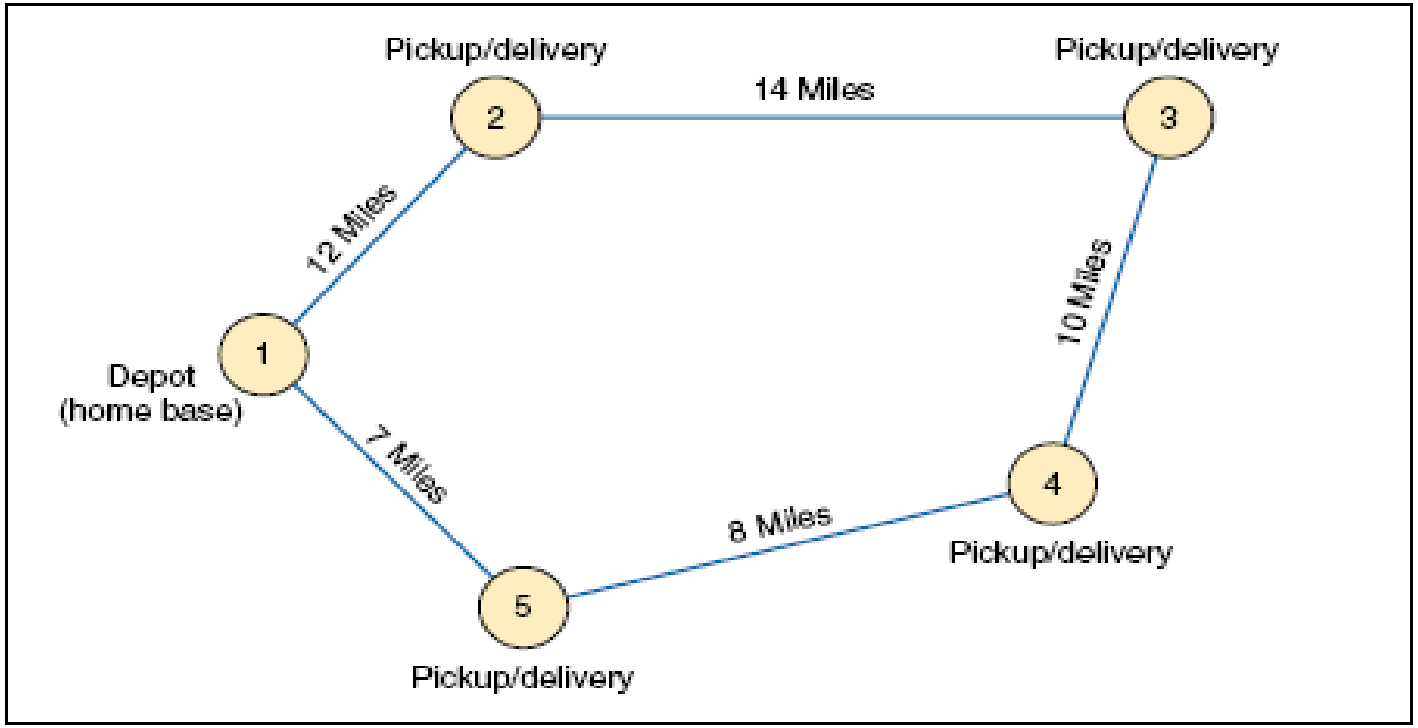
# Objectives and Constraints of Routing and Scheduling Problems

- Minimize total cost of service delivery or route by one or more of the following:
  - Minimize total distance traveled
  - Minimize total time traveled
  - Minimize number of vehicles utilized
  - Minimize number of personnel on duty
- Subject to one of more constraints (not exhaustive):
  - Vehicles
    - ✓ Capacities
    - ✓ Release Time, Maximum Time and Down Time
  - Customers
    - ✓ Time windows (Timing open for receiving of goods)
    - ✓ Priority
    - ✓ Pickup and delivery

# Characteristics of Routing and Scheduling Problems

- Can be represented using network diagrams, using nodes and arcs
  - Nodes represent locations to be visited. Depot node is the begin/end location
  - Arcs represent the possible paths among the locations. Arcs can be directed (one-way, represented by arrow) or undirected (two ways, represented by line)

# Feasibility of a Tour




Example of a tour for the above network: 1 → 2 → 3 → 4 → 5 → 1

**For a tour to be feasible,**

1. It must start and end with the depot node
2. All nodes must be included
3. A node must be visited only once

# Classification of Routing and Scheduling Problems

Routing and scheduling problems can be classified into the following categories:

- **Traveling salesman problem**
- Multiple traveling salesman problem
- **Vehicle routing problem**
- Vehicle routing problem with time windows
- Pickup and delivery problem with time windows

Will be covered in more details in the next E-Learning lesson!

# Solution approaches to the Routing and Scheduling Problems

- Exact algorithms such as integer programming algorithms
  - For a problem with *n* drop-off points, there are *n*! possible routings. E.g. 8 drop-off points → 40320 possible routings!
  - The solution space increases extremely fast as the number of drop-off points increases, and therefore exact algorithms are often not suitable for large size problems.

- **Heuristic algorithms**
  - Rule-of-thumb methods
  - May not achieve optimality but solutions are often reasonably good.

# Solution approaches to the Routing and Scheduling Problems

- Commonly used vehicle routing heuristics
  - **Construction algorithm:** an algorithm that determines a tour according to some construction rules. Examples are **Nearest Neighbor method and Clarke and Wright Savings method.**
  - **Improvement algorithm:** an algorithm that performs a sequence of arc or node exchanges within or between vehicle routes to improve the current tour.
  - **Two-phase algorithm:** an algorithm that constructs vehicle routes in two phases:
    - ✓ **Cluster-first-route-second method:** customers are first organized into feasible clusters, and a vehicle route is constructed for each of the clusters.
    - ✓ **Route-first-cluster-second method:** a tour is first built on all customers and the tour is then segmented into feasible vehicle routes.

# Nearest Neighbour Method

# Nearest Neighbor Method

1. Select a node (e.g. node 1) to start with

2. Add the closest node to node 1

3. Repeat by adding to the last node the closest unvisited node until no more nodes are available

4. Connect the last node with the first node

# Nearest Neighbor Method – An Example

| From | To (Distance in km) | | | | | | |
|---|---|---|---|---|---|---|---|
| | A (Depot) | B | C | D | E | F | G |
| A (Depot) | 0 | 22 | 19 | 15 | 7 | 12 | 26 |
| B | 22 | 0 | 5 | 16 | 18 | 12 | 30 |
| C | 19 | 5 | 0 | 6 | 14 | 8 | 16 |
| D | 15 | 16 | 6 | 0 | 8 | 3 | 21 |
| E | 7 | 18 | 14 | 8 | 0 | 6 | 25 |
| F | 12 | 12 | 8 | 3 | 6 | 0 | 28 |
| G | 26 | 30 | 16 | 21 | 25 | 28 | 0 |

Starting from A (Depot), nearest node is E.
From E, the nearest node is F.
From F, the nearest node is D.
And so on…

Tour determined using Nearest Neighbour method:
A -> E -> F -> D -> C -> B -> G -> A
Total distance = 7+6+3+6+5+30+26 = 83 km

# Clarke & Wright Savings Method

# Clarke and Wright Savings Method

1. Select a depot node (denote as node 1)
2. Compute the savings $S_{ij}$ , for linking nodes *i* and *j*

$$S_{ij} = c_{1i} + c_{1j} - c_{ij} \text{ for } i \text{ and } j = \text{nodes } 2,\ 3,\ \ldots,\ n.$$

where $c_{ij} = \text{the cost of travelling from node } i \text{ to node } j.$

3. Rank the savings from the largest to the smallest

4. Starting from the top of the list, form larger subtours by linking **'appropriate'** nodes *i* and *j* (so that *j* is visited immediately after *i* on the resulting route)

Stop when a complete tour is formed.

'appropriate' means:

- Not violating route constraints such as vehicle capacity or time window constraints; **AND**
- *i* and *j* fulfil one of the following conditions:

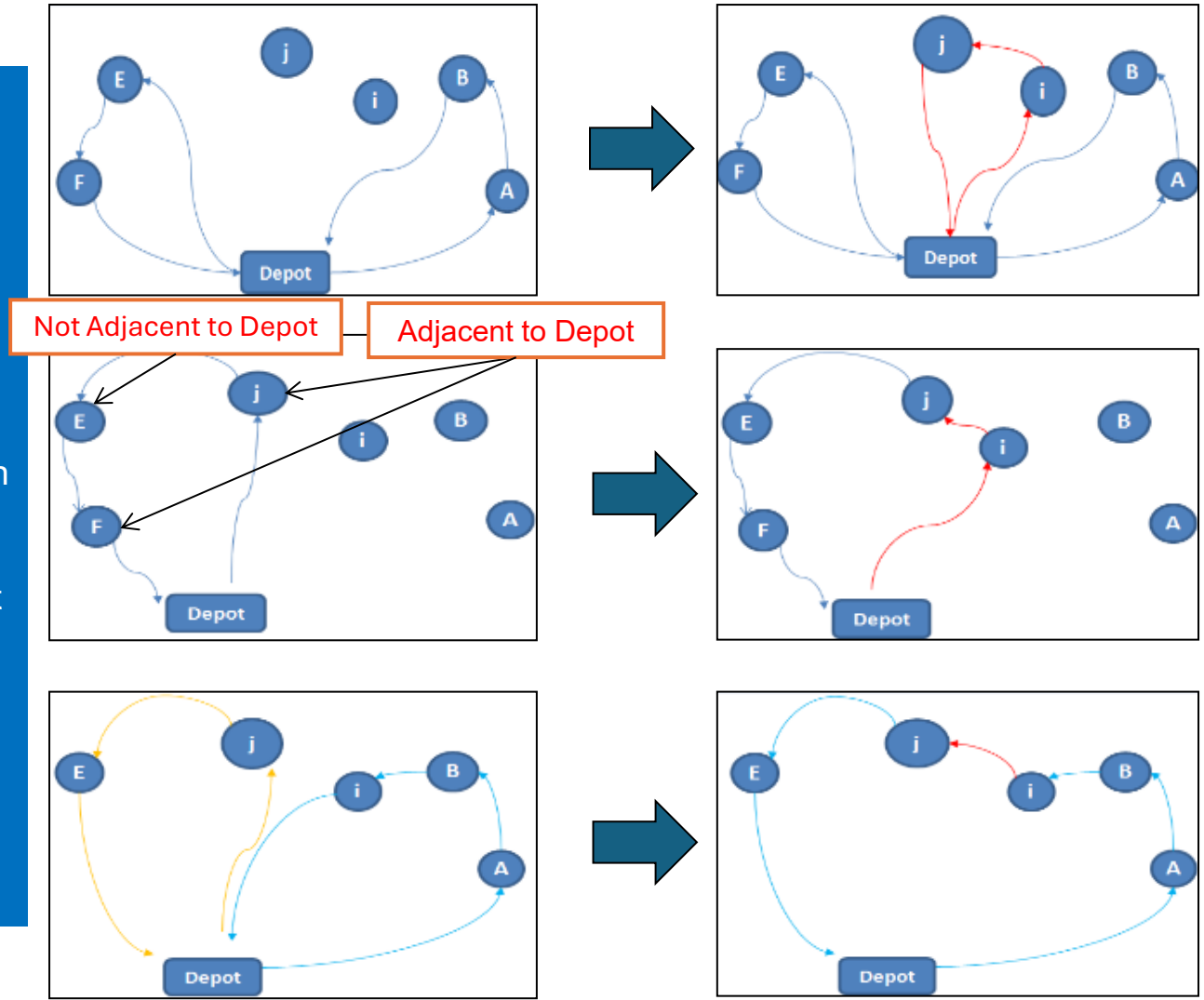
# Clarke and Wright Savings Method – Tour Construction

➢ ***i* and *j* fulfil one of the following conditions:**

A. Neither i nor j has already been assigned to a route. Initiate a new route including both i and j

B. Exactly *one* of the two points (i or j) has already been included in an existing route and that point is adjacent to the depot. Add link (i, j) to that same route.

C. Both i and j have already been included in two different existing routes and both of them are adjacent to the depot. Merge the two routes.



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# C&W Savings Calculation

## Distance Table

| From | To (Distance in km) | | | | | | |
|---|---|---|---|---|---|---|---|
| | A (Depot) | B | C | D | E | F | G |
| A (Depot) | 0 | 22 | 19 | 15 | 7 | 12 | 26 |
| B | 22 | 0 | 5 | 16 | 18 | 12 | 30 |
| C | 19 | 5 | 0 | 6 | 14 | 8 | 16 |
| D | 15 | 16 | 6 | 0 | 8 | 3 | 21 |
| E | 7 | 18 | 14 | 8 | 0 | 6 | 25 |
| F | 12 | 12 | 8 | 3 | 6 | 0 | 28 |
| G | 26 | 30 | 16 | 21 | 25 | 28 | 0 |

$S_{BD} = D_{AB} + D_{AD} - D_{BD}$ --> $S_{BD} = 22 + 15 - 16 = 21$

## Clarke and Wright Savings Table

| From | Savings | | | | | | |
|---|---|---|---|---|---|---|---|
| | A (Depot) | B | C | D | E | F | G |
| A (Depot) | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| B | 0 | 0 | 36 | 21 | 11 | 22 | 18 |
| C | 0 | 36 | 0 | 28 | 12 | 23 | 29 |
| D | 0 | 21 | 28 | 0 | 14 | 24 | 20 |
| E | 0 | 11 | 12 | 14 | 0 | 13 | 8 |
| F | 0 | 22 | 23 | 24 | 13 | 0 | 10 |
| G | 0 | 18 | 29 | 20 | 8 | 10 | 0 |

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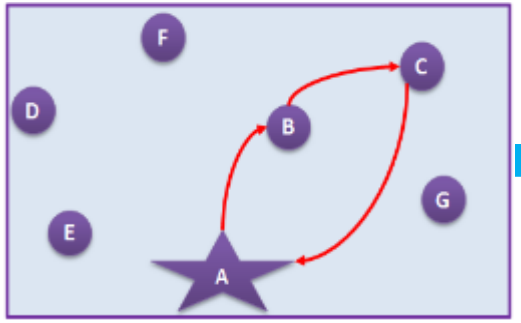
# C&W Savings Tour Construction

| Rank(Dist.) | From | To | Path | |
|---|---|---|---|---|
| 1(36) | B | C | A - B - C - A | Initiate new tour |
| 2(29) | C | G | A - B - C - G - A | Join C & G |
| 3(28) | C | D | - | C is not adjacent to A (Depot) |
| 4(24) | D | F | A - D - F - A, A - B - C - G - A, | Initiate new tour |
| 5(23) | C | F | - | C is not adjacent to A (Depot) |
| 6(22) | B | F | A - D - F - B - C - G - A, | Join B & F; Merge the 2 tours |
| 7(21) | B | D | - | B, D and G are already assigned to the same route |
| 8(20) | D | G | - | |
| 9(18) | B | G | - | |
| 10(14) | D | E | A - E - D - F - B - C - G - A, End. | Join D & E |
| 11(13) | E | F | | |
| 12(12) | C | E | | |
| 13(11) | B | E | | |
| 14(10) | F | G | | |
| 15(8) | E | G | | |

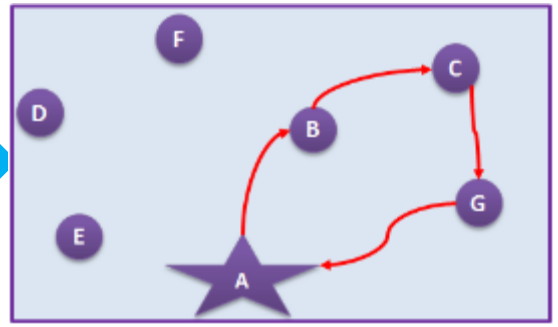
**Route Constructed: A - E - D - F - B - C - G - A**

**Total Distance**
**= 7 + 8 + 3 + 12 + 5 + 16 + 26 = 77 km**

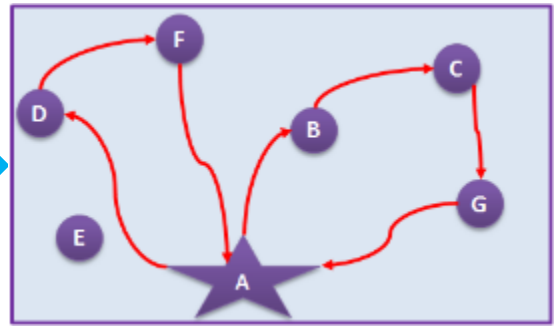
# C&W Savings Method – Tour Construction



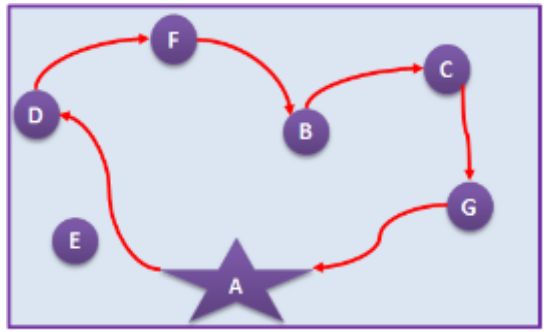

Initiate a new route: A-B-C-A



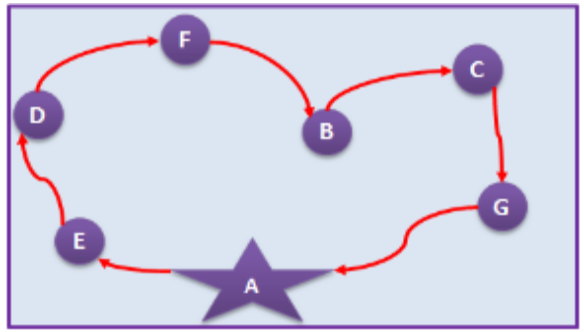

Link G to C: A-B-C-G-A




Initiate another route: A-D-F-A




Join both routes by linking B and F: A-D-F-B-C-G-A




Link E to D: A-E-D-F-B-C-G-A

| Route (s) | Distance |
|---|---|
| A - E - D - F - B - C - G - A | 77 |

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# C&W Savings Method with Capacity Constraint

**Customer demand**

| Location | B | C | D | E | F | G |
|---|---|---|---|---|---|---|
| Demand | 20 | 19 | 15 | 25 | 18 | 22 |

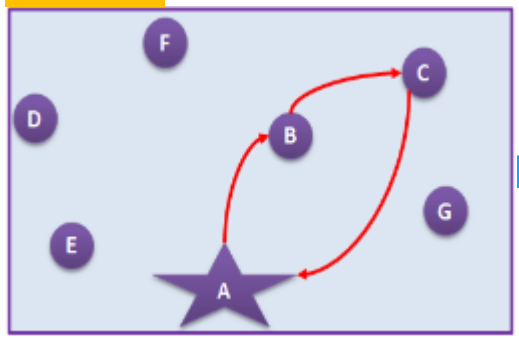
| Rank (Dist.) | From | To | Current Tour | Cumulative Load | Remark |
|---|---|---|---|---|---|
| 1(36) | B | C | A - B - C - A | 20 + 19 = 39 | Initiate a new route: A-B-C-A |
| 2(29) | C | G | A - B - C - G - A | 20 + 19 + 22 = 61 | Link G with C |
| 3(28) | C | D | - | - | C is not adjacent to A (Depot) |
| 4(24) | D | F | A - D - F - A, A - B - C - G - A | 15 + 18 = 33, 20 + 19 + 22 = 61 | Initiate another route: A-D-F-A |
| 5(23) | C | F | - | - | C is not adjacent to A (Depot) |
| 6(22) | B | F | A - D - F - B - C - G - A | 15 + 18 + 20 + 19 + 22 = 94 | Link B and F. |
| 7(21) | B | D | - | - | Both B and D are already in a route |
| 8(20) | D | G | - | - | Both D and G are already in a route |
| 9(18) | B | G | - | - | Both B and G are already in a route |
| 10(14) | D | E | A - D - F - B - C - G - A, A - E - A | 15 + 18 + 20 + 19 + 22 = 94, 25 | Cannot link D and E, because truck capacity (100 boxes) will be violated, start new route for E. End |
| 11(13) | E | F | | | |
| 12(12) | C | E | | | |
| 13(11) | B | E | | | |
| 14(10) | F | G | | | |
| 15(8) | E | G | | | |

Truck 1: A - D - F - B - C - G - A with load 94 boxes and total distance 77km
Truck 2: A - E - A with load 25 boxes and total distance 14km

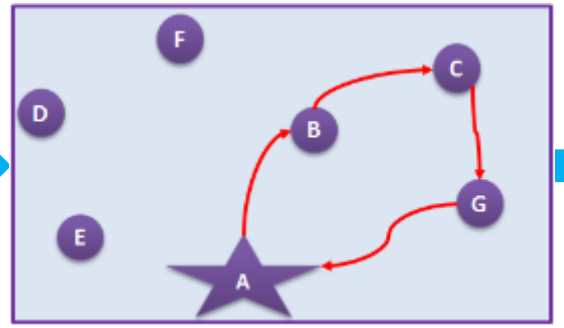
Vehicle capacity = 100 boxes

# C&W Savings Method with Capacity Constraint

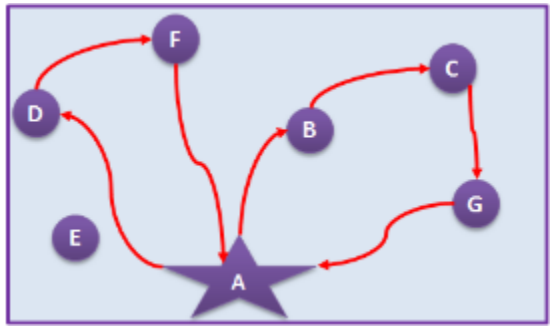
**Tour 1**



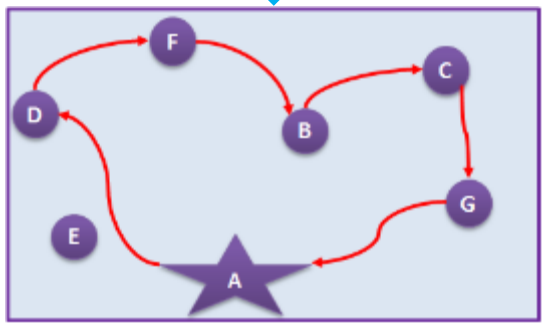

Initiate a new route: A-B-C-A



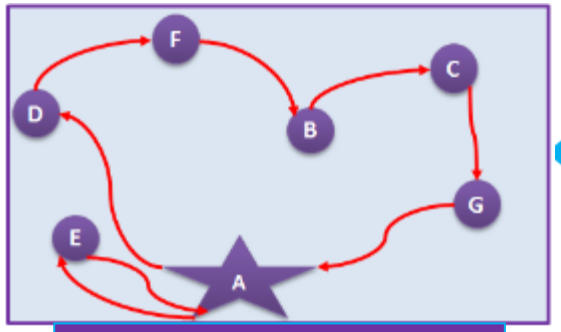

Link G to C: A-B-C-G-A




Initiate another route: A-D-F-A




**Tour 2**




Initiate new route: A-E-A

| Route (s) | Demand (boxes) | Distance |
|---|---|---|
| A - D - F - B - C - G - A | 94 | 77 |
| A - E - A | 25 | 14 |
| | Total Distance | 91 km |

# Summary

- Routing and scheduling problems can be found in our daily lives and operations.
- Objective is to find the shortest tours that cover all the places (nodes) without a repeat visit and begin / end at one node.
- Heuristic methods can be used to provide feasible solutions to complex routing and scheduling problems.
- Clarke & Wright Savings method usually perform better than the Nearest Neighbour method; both approaches do not guarantee an optimal solution.

# Learning Outcomes

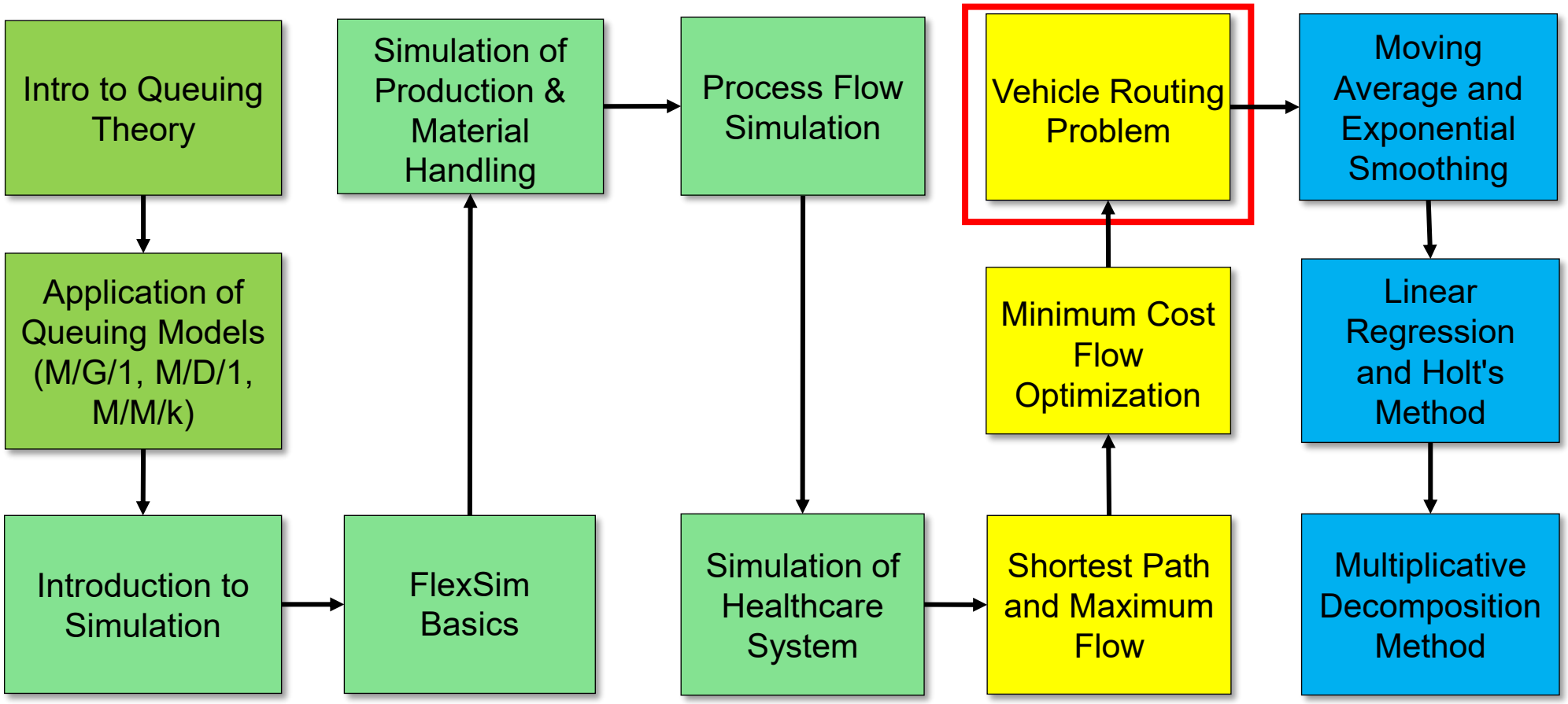
At the end of the lesson, students should be able to:

- Identify what is a Vehicle Routing and Scheduling problem: Objectives, constraints, characteristics and classification of problems.
- Apply heuristic algorithm: Nearest Neighbour Method to obtain a feasible route.
- Apply heuristic algorithm: Clarke and Wright Savings method to obtain a feasible route.

# Quiz

Click the **Quiz** button to edit this object

Which statement is false?

- For a tour to be feasible, it must start and end at the same node.
- Routing involves forming the sequence of visiting the locations.
- Scheduling involves planning the timing for each location to be visited.
- Routing problems are frequently solved using Linear Programming method.
- The aim of route planning is to minimize the overall cost of providing the service.

# E211 Progression